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30 January 1981
SUPERSEDING
MIL-C-29140(SA)
20 January 1975

MILITARY SPECIFICATION

CLOTH, BASKET WEAVE, POLYESTER AND COTTON (CREASE-RESISTANT FINISH)

*This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 Scope This specification covers the requirements for two types of polyester and cotton blend fabrics treated with a crease resistant finish.

Type I - 4.8 oz/yd^2
Type II - 4.3 oz/yd^2

2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered
MIL-T-43624 - Thread, Polyester, Spun

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

3. REQUIREMENTS

* 3.1 Standard sample The finished cloth shall match the standard sample for shade, uniformity of shade (see 3.5.4), and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

*3.2 First article When specified (see 6.2), the contractor shall furnish a sample of cloth for first article inspection and approval.

3.3 Materials

3.3.1 Fiber blend

3.3.1.1 Polyester fiber The fiber shall be polyethylene glycol terephthalate, semi-dull, when tested as specified in 4.5.

3.3.1.2 Cotton fiber The cotton shall be carded and combed, when tested as specified in 4.5.

* 3.3.2 Yarns The warp and filling yarns shall be a blend of 65 (± 5) percent polyester and the remaining percentage cotton based on the dry weight of the desized specimens before finish treatment, when tested as specified in 4.5. The fibers shall be spun into singles yarn for both the warp and filling.

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* 3.4 Physical requirements The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical requirements

Type	Weight (g/m^2)	(oz/yd^2) (Min)	Yarns per inch (cm) (Min.) warp	filling)	Breaking strength (lbs)(newtons)(Min) warp	filling	Tearing strength (lbs)(newtons)(Min) warp	filling
I	4.8 (162.7)		92 (36.2)	72 (28.4)	110 (489.5)	75 (333.8)	9.5 (42.3)	8.5 (37.8)
II	4.3 (145.8)		97 (38.2)	55 (21.7)	125 (556.3)	70 (311.5)	11.5 (51.2)	6.0 (26.7)

3.4.1 Width The width of the cloth shall be as specified (see 6.2), and shall be the minimum width, inclusive of selvage.

3.4.2 Weave The weave shall be a modified basket weave as shown in Figure 1.

* 3.5 Color The color of the finished cloth shall be as specified (see 6.2) and shall match the standard sample (see 6.3).

3.5.1 Sulfur content The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5.

* 3.5.2 Matching The finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.5.3 Colorfastness The finished cloth shall show fastness to light, laundering (after three cycles), perspiration and crocking equal to or better than the standard sample.

3.5.4 Uniformity of shade The uniformity of shade shall be equal to or better than that displayed by the standard sample.

3.6 Nonfibrous material The total nonfibrous material content of resin finished cloth shall not exceed 6.0 percent when tested as specified in 4.5.

3.7 pH The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

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3.8 Dimensional stability The finished cloth shall not shrink nor elongate more than 2.0 percent in either the warp or the filling when tested as specified in 4.5. The pre-shrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.9 Seam efficiency The finished cloth shall have a seam efficiency of not less than 90 percent when tested as specified in 4.5.

* 3.10 Finish The cloth shall be desized, scoured, dyed, heat set, treated with a resin and cured to provide a crease-resistant finish and hand equal in characteristic to that of the standard sample.

3.11 Length and put-up Unless otherwise specified (see 5.1.1 and 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.6m). Each length shall be put-up in full width rolls as specified in PPP-P-1134 with the face inward.

3.12 Marking The face of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.13 Fiber identification Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Product Identification Act.

*3.14 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection When required, the first article of finished cloth shall be visually inspected for color, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

*4.4 Quality conformance inspection Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

*4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

*4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4m) the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m) only 100 yards (91.4m) shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) (0.9m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "C" for slubs and "B" for knots, shall be scored. No linear yard (increments of 1 yard (0.9m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards (83.6m) of two or more individual rolls exceed 38.0 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls exceed the value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

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*4.4.2.1.1 Demerit points Demerit points shall be assigned as follows:

For defects 3 inches (7.6cm) or less in any dimension	- one point
For defects exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension	- two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.9cm) in any dimension	- three points
For defects exceeding 9 inches (22.9cm) in any dimensions	- four points

The following defects, when present, shall be scored four points for each yard (0.9m) in which they occur.

Baggy, ridgy or wavy cloth
Width less than specified
Characteristics of finish, hand and shade not equal to the standard sample.
Non-uniformity of shade (poor dyed penetration, mottled, streaky or cloudy).
Excessive neppiness.

*4.4.2.2 Examination for length

*4.4.2.2.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

*4.4.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

*4.4.2.3 Examination for shade During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off-shade (compared to the standard sample and applicable shade range), shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

*4.4.2.4 Examination for face identification, identification of preshrinking process and non-conformance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinking process by name or trademark on the cloth or ticket, or two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

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*4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

*4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing final results. The sample unit shall be 4 1/2 yards (4.1m), full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size yards	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test methods

Characteristic	Requirement Paragraph	Test Method
Fiber identification	3.3.1	1/
Fiber content:		
Polyester (percent)	3.3.2	2100, 4.5.1
Cotton (percent)	3.3.2	2100, 4.5.1
Yarn (singles);		
Warp	3.3.2	Visual 2/
Filling	3.3.2	Visual 2/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Tear strength:		
Warp	3.4	5132
Filling	3.4	5132
Weave	3.4.2	Visual 2/
Sulfur content	3.5.1	2020

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Table II - Test methods (cont'd)

Characteristic	Requirement Paragraph	Test Method
Colorfastness:		
Light	3.5.3	5660
Laundering (after 3 cycles)	3.5.3	5610 5/ 6/
Perspiration	3.5.3	5680
Crocking	3.5.3	5651
Nonfibrous materials	3.6	2610
pH	3.7	2811
Dimensional stability	3.8	5556 3/
Seam efficiency	3.9	5110 4/
Finish	3.10	1/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.
2/ One determination per sample unit and the results reported as "pass" or "fail".
3/ Cotton laundering procedure shall be followed.
4/ The needle shall measure 0.040 inch (+ 0.001) across the blade at the eye. The thread shall be either polyester/cotton in accordance with MIL-T-43548, ticket No. 50, 2 ply for the needle and ticket No. 70, 2 ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or 2, ticket No. 40, 3 ply for the needle and ticket No. 70, 2 ply for the looper.
5/ The specimens must be dried after each of the 3 laundering cycles.
6/ Only the stain on the cotton and polyester fibers of the color transfer cloth shall be evaluated.

*4.5.1 Polyester and cotton fiber content The general procedures of method 2100 shall be followed with the fiber content percentages calculated as follows:

$$\text{Polyester, percent} = \frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}}$$

Percent cotton = 100 - percent polyester

*4.5.1.1 Report Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

5. PACKAGING

5.1 Packaging Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C The cloth, put up as specified (see 3.11), shall be packaged in accordance with the applicable requirements of PPP-P-1134.

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5.2 Packing Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

*6.1 Intended use The cloth covered by this specification is intended for use in the blue utility shirt worn by Navy and Coast Guard personnel.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.2).
c. Type and color required (see 3.4 and 3.5).
d. Width required (see 3.4.1).
e. Length required if other than specified (see 3.11).
f. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Standard5sample For access to the standard sample, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Sears Fabric Defect Scales are available from Sears Roebuck and Company Department, 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, ILL 60684.

*6.5 Recycled material It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

*6.6 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
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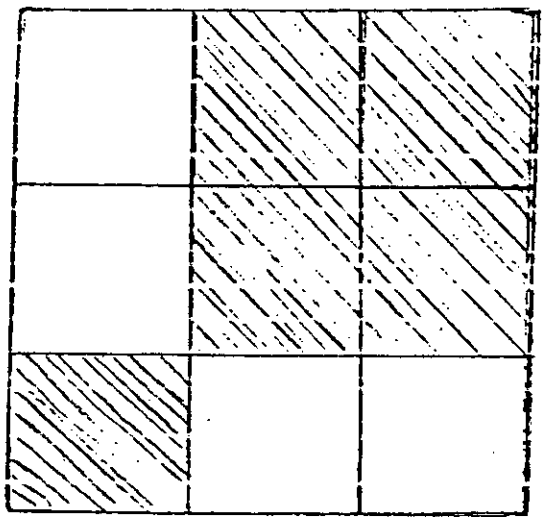


FIGURE 1

Weave Diagram

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